

Chapter 45: The matrix of a linear transformation Worksheet

Recall: $M_{DB}(T)$ denotes the matrix representation of the linear transformation $T : V \rightarrow W$ using ordered bases B for V and D for W . $M_B(T)$ denotes the matrix representation of the linear operator T with respect to the ordered basis B .

1. Are each of the following statements true or false?

- (a) Let $T : V \rightarrow W$ be a linear transformation where $\dim V = n$ and $\dim W = m$, and let B and D be ordered bases of V and W . Then $M_{DB}(T)$ is an $n \times m$ matrix.
- (b) Let $T : V \rightarrow W$ be a linear transformation. If B_1 and D_1 are ordered bases of V and W , and B_2 and D_2 are another ordered bases of V and W , then $M_{D_1B_1}(T) = M_{D_2B_2}(T)$.
- (c) Let $T : V \rightarrow W$ be a linear transformation. If B_1 and D_1 are ordered bases of V and W , and B_2 and D_2 are another ordered bases of V and W , then the ranks of $M_{D_1B_1}(T)$ and $M_{D_2B_2}(T)$ are the same.
- (d) The matrix of the zero transformation $0 : V \rightarrow W$ corresponding to any bases is the zero matrix.
- (e) If $T : P_3 \rightarrow P_3$ is a linear operator with matrix representation $M = M_B(T)$, and the rank of M is 3, then T is an isomorphism.

2. Suppose $T : P_2 \rightarrow M_{22}$ is a linear transformation represented by $M_{DB}(T) = \begin{bmatrix} -2 & 2 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$

with $B = (1, x^2, x)$ and $D = \left(\begin{bmatrix} -3 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \right)$. Find $T(2x^2 - x + 3)$

(Note: your final answer needs to be a 2×2 matrix).

3. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T \left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \right) = \begin{bmatrix} u_1 + 2u_2 \\ 2u_1 - u_2 \end{bmatrix}$. Let $B = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ be the standard basis for $\mathbb{R}^2$ and let $D = \left\{ \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\}$. Find the matrices of T corresponding to

(a) B , i.e. $M_B(T) = M_{BB}(T)$.

(b) D , i.e. $M_D(T) = M_{DD}(T)$.

(c) Find $T \left(\begin{bmatrix} 1 \\ 2 \end{bmatrix} \right)$ using the definition of T and also using the matrices found in part (a) and (b).

4. Let $T : V \rightarrow V$ be the linear transformation defined by $T(\mathbf{v}) = c\mathbf{v}$ for all $\mathbf{v} \in V$; here $\dim V = n$ and c is some fixed constant. Show that the matrix $M_{BB}(T)$ corresponding to any ordered basis B for V is the scalar matrix cI_n .

5. Let $T : P_2 \rightarrow P_2$ be a linear transformation with $M_B(T) = \begin{bmatrix} 1 & 3 & 5 \\ -2 & 1 & -3 \\ 3 & 2 & 8 \end{bmatrix}$ with respect to the ordered basis $B = (x + 1, x^2, 2 - x)$. Find a basis for the kernel of T . (Note: your answer(s) need to be elements of P_2).

Hint: Start by finding a basis for the nullspace of $M_B(T)$.